Plot Papier

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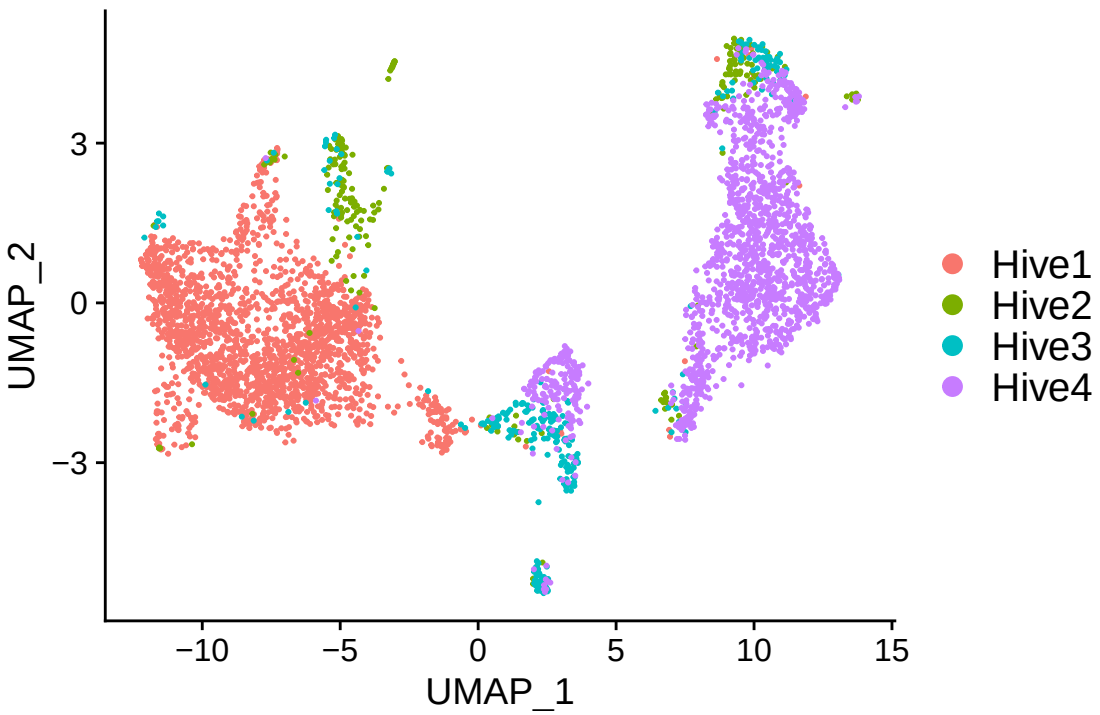
Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(RColorBrewer))
suppressMessages(library(dittoSeq))
suppressMessages(library(ComplexHeatmap))
suppressMessages(library(plotly))
suppressMessages(library(tidyHeatmap))
```

Loading data

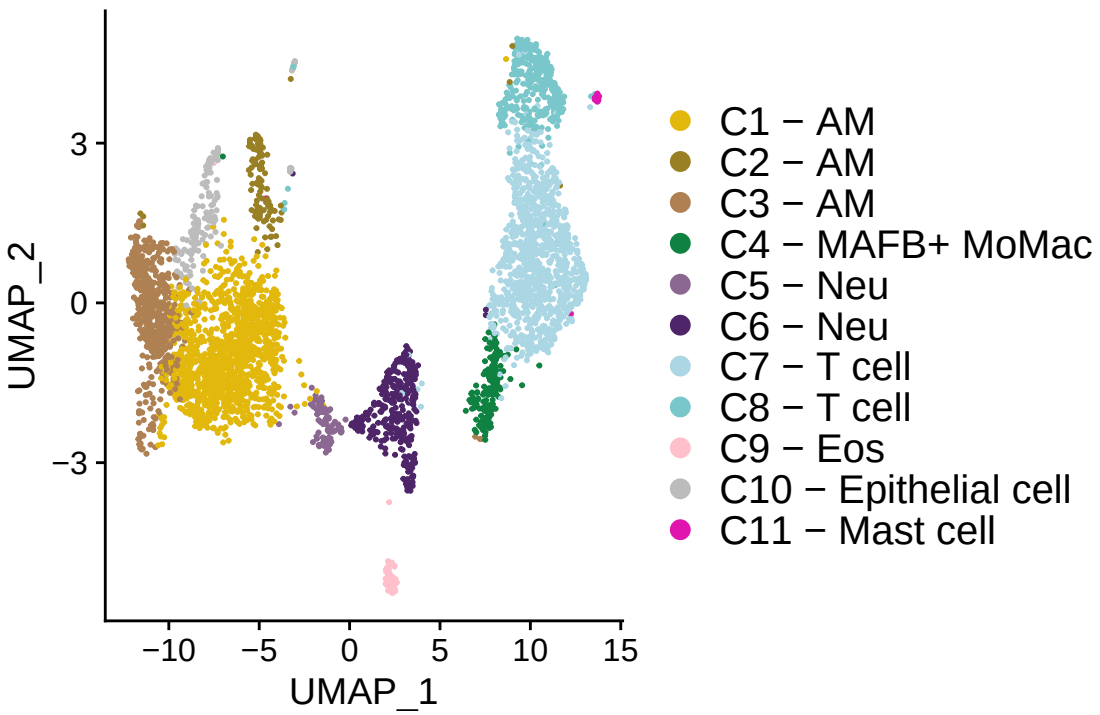
```
Cell_HiveAnnotate <- readRDS("../Hive-1-Processing/Lavage_cellsHive.rds")
```

```
DimPlot(Cell_HiveAnnotate, reduction = "umap", group.by = "orig.ident")+
theme(legend.text   = element_text(size = 15))+ labs(title = NULL)
```




```
#ggsave("Umap_PerSample.pdf", width = 8, height = 4)
```

```
Cell_HiveAnnotate$Annotation <- Cell_HiveAnnotate$seurat_clusters
levels(Cell_HiveAnnotate$Annotation) <- c("C1 - AM", "C7 - T cell", "C3 - AM", "C8 - T
cell", "C6 - Neu", "C4 - MAFB+ MoMac", "C10 - Epithelial cell", "C2 - AM", "C5 - Neu",
"C9 - Eos", "C11 - Mast cell")

Cell_HiveAnnotate$Annotation <- factor(Cell_HiveAnnotate$Annotation, levels =  c("C1 -
AM", "C2 - AM", "C3 - AM", "C4 - MAFB+ MoMac", "C5 - Neu", "C6 - Neu", "C7 - T cell", "C8
- T cell", "C9 - Eos", "C10 - Epithelial cell", "C11 - Mast cell"))

# Je modifie les couleurs pour qu'elles soient égalent à celle du HIVE
col <- c("#E2B80C", "#998025", "#AE8052", "#0F8140", "#8A6892", "#4F2569", "#ABD6E4",
"#79C7CB", "#FFC0CB", "#BDBCBC", "#DF15AE")

DimPlot(Cell_HiveAnnotate, group.by = "Annotation", cols = col)+ theme(legend.text   =
element_text(size = 15)) + labs(title = NULL)
```




```
#ggsave("../plot_paper/Umap_Annotation.pdf", width = 8, height = 4)
```

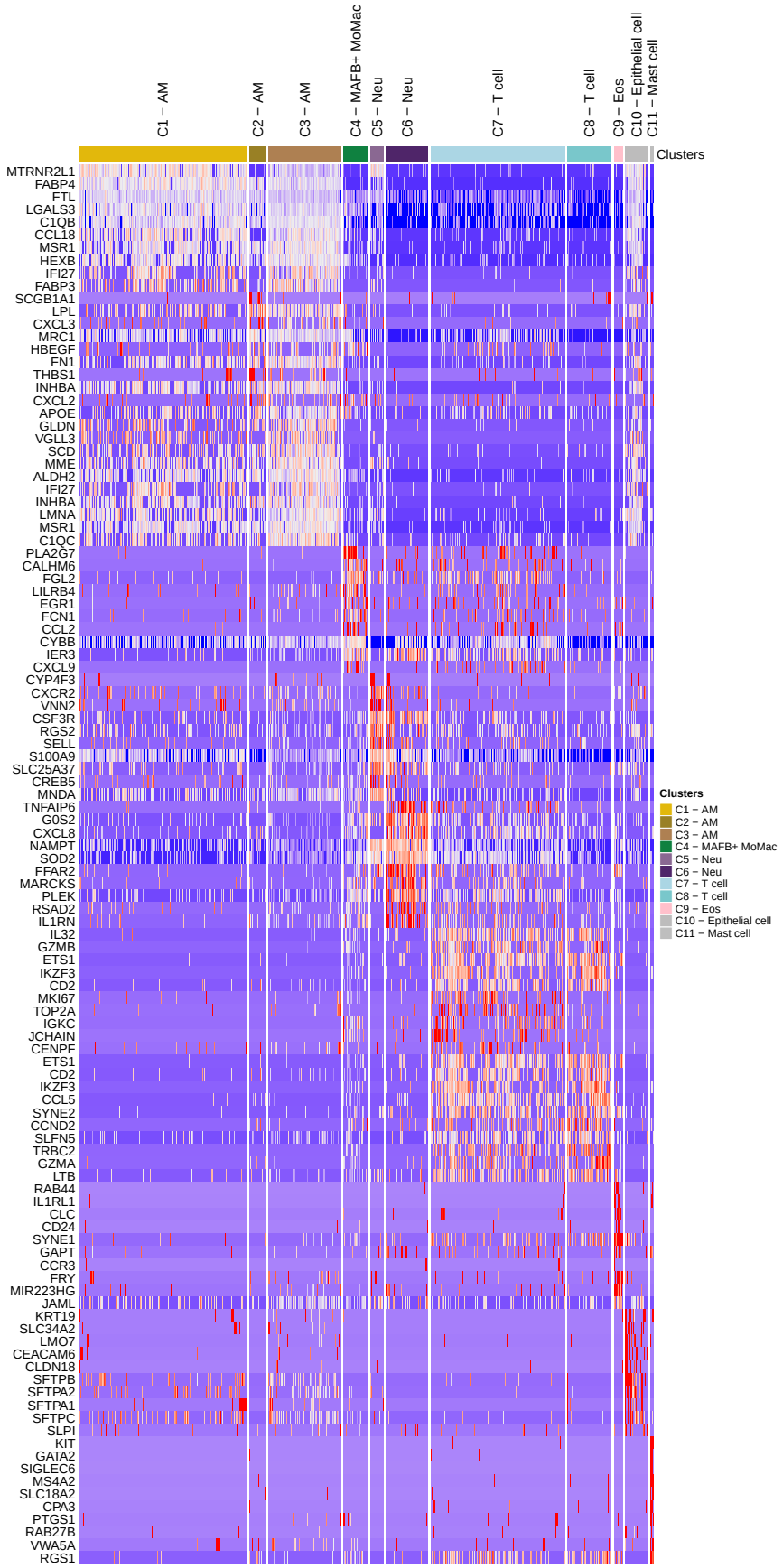
Differential Expression Analysis

```
Idents(Cell_HiveAnnotate) <- "Annotation"
lavage.markers <- FindAllMarkers(Cell_HiveAnnotate, only.pos = TRUE, min.pct = 0.25)

lavage.markers %>%
    group_by(cluster) %>%
    top_n(n = 10, wt = avg_log2FC) -> top10

mat <- as.matrix(GetAssayData(object = Cell_HiveAnnotate, slot =
"data")[as.character(top10$gene),])

df <- as.data.frame(Cell_HiveAnnotate$Annotation)
colnames(df) <- "Clusters"
color_df <- list(Clusters =
                    c("C1 - AM" = "#E2B80C",
                      "C2 - AM" = "#998025",
                      "C3 - AM" = "#AE8052",
                      "C4 - MAFB+ MoMac" = "#0F8140",
                      "C5 - Neu" = "#8A6892",
                      "C6 - Neu" = "#4F2569",
                      "C7 - T cell" = "#ABD6E4",
                      "C8 - T cell" = "#79C7CB",
                      "C9 - Eos" = "#FFC0CB",
                      "C10 - Epithelial cell" = "#BDBCBC",
                      "C11 - Mast cell" = "gray"))
```

```r
Heatmap <- Heatmap(t(scale(t(mat))), show_column_names = F,
        column_split = Cell_HiveAnnotate$Annotation,
        cluster_column_slices = F,
        cluster_rows = F,
        top_annotation = HeatmapAnnotation(df = df, col = color_df),
        use_raster = F,
        show_heatmap_legend = F,
        show_column_dend = F,
        column_title_rot = 90,
        row_names_side = "left")

#save_pdf(Heatmap, "Heatmap_Hive.pdf",width = 30, height = 45, units ="cm")
```

Heatmap top genes per clusters

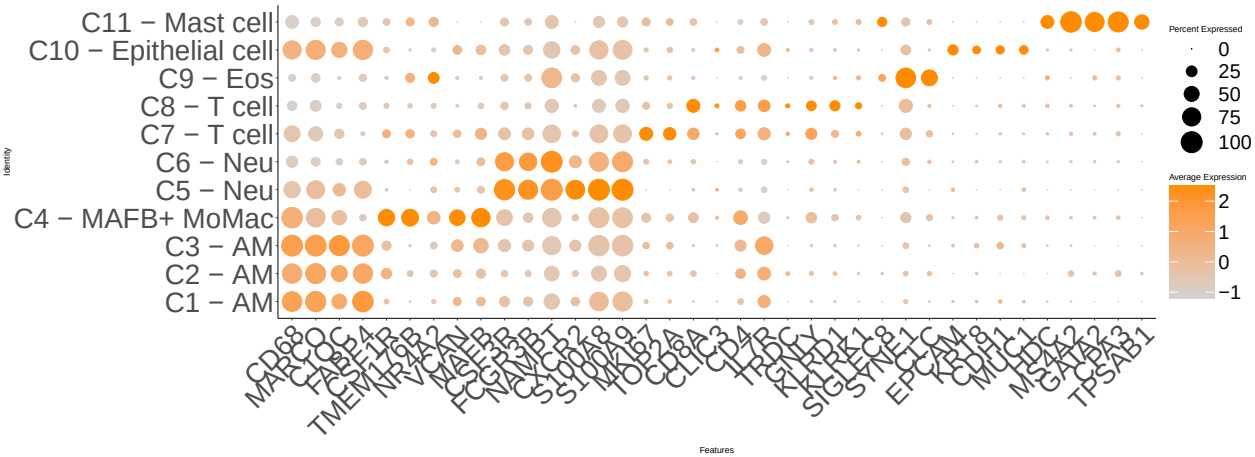
```r
Heatmap
```


C1 – AM
C2 – AM
C3 – AM
C4 – MAFB+ MoMac
C5 – Neu
C6 – Neu
C7 – T cell
C8 – T cell
C9 – Eos
C10 – Epithelial cell
C11 – Mast cell
Clusters
MTRNR2L1
FABP4
FTL
LGALS3
C1QB
CCL18
MSR1
HEXB
IFI27
FABP3
SCGB1A1
LPL
CXCL3
MRC1
HBEGF
FN1
THBS1
INHBA
CXCL2
APOE
GLDN
VGLL3
SCD
MME
ALDH2
IFI27
INHBA
LMNA
MSR1
C1QC
PLA2G7
CALHM6
FGL2
LILRB4
EGR1
FCN1
CCL2
CYBB
IER3
CXCL9
CYP4F3
CXCR2
VNN2
CSF3R
RGS2
SELL
S100A9
SLC25A37
CREB5
MNDA
TNFAIP6
G0S2
CXCL8
NAMPT
SOD2
FFAR2
MARCKS
PLEK
RSAD2
IL1RN
IL32
GZMB
ETS1
IKZF3
CD2
MKI67
TOP2A
IGKC
JCHAIN
CENPF
ETS1
CD2
IKZF3
CCL5
SYNE2
CCND2
SLFN5
TRBC2
GZMA
LTB
RAB44
IL1RL1
CLC
CD24
SYNE1
GAPT
CCR3
FRY
MIR223HG
JAML
KRT19
SLC34A2
LMO7
CEACAM6
CLDN18
SFTPB
SFTPA2
SFTPA1
SFTPC
SLPI
KIT
GATA2
SIGLEC6
MS4A2
SLC18A2
CPA3
PTGS1
RAB27B
VWA5A
RGS1
Clusters
C1 – AM
C2 – AM
C3 – AM
C4 – MAFB+ MoMac
C5 – Neu
C6 – Neu
C7 – T cell
C8 – T cell
C9 – Eos
C10 – Epithelial cell
C11 – Mast cell

Dotplot of relevant markers

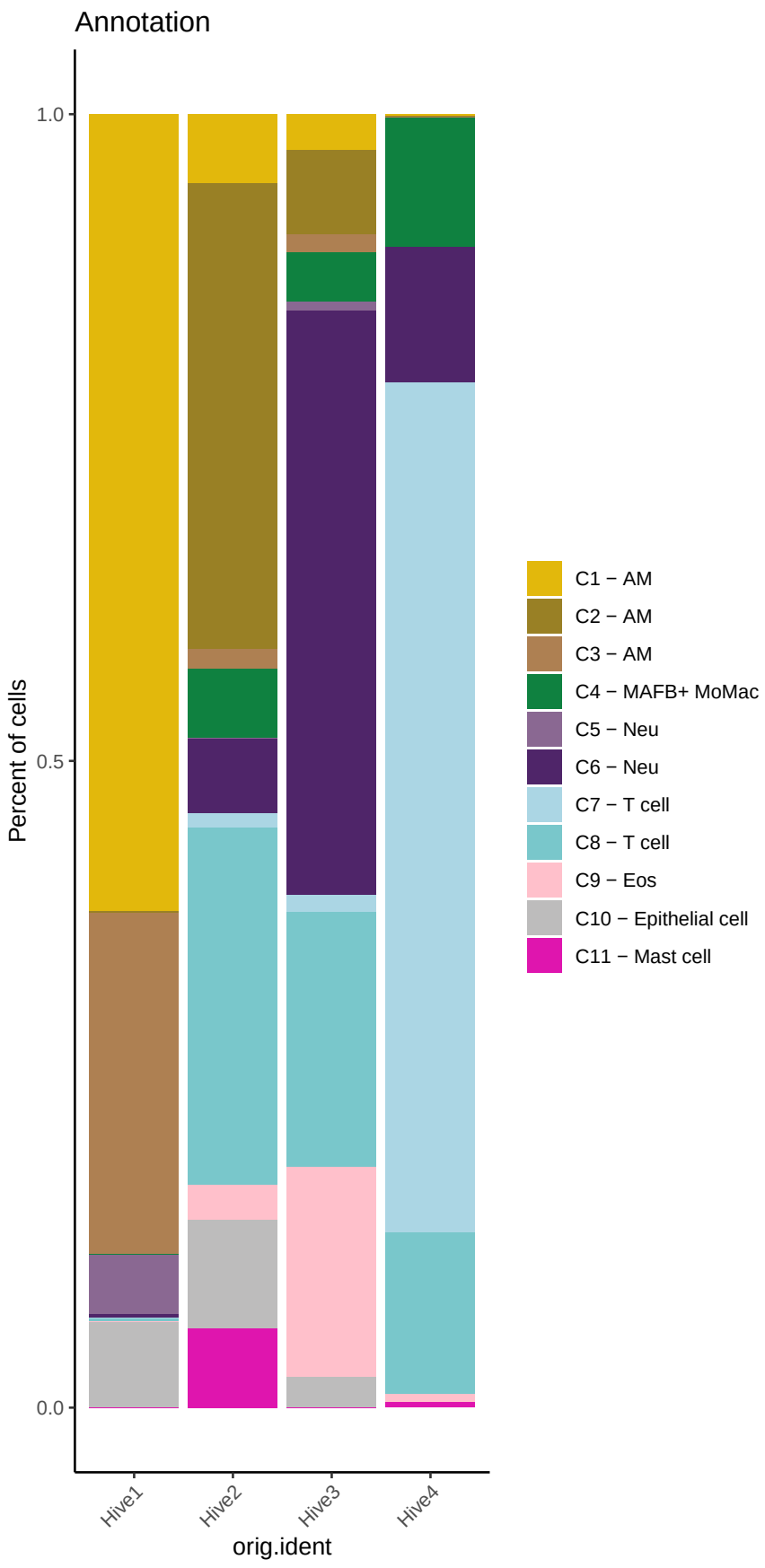
```
markers_neutro <- c("CSF3R","FCGR3B", "NAMPT", "CXCR2", "S100A8", "S100A9")
marker_epi <- c("EPCAM", "KRT8", "CDH1", "MUC1")
Marker_NK <- c("TRDC", "GNLY", "KLRD1", "KLRK1")
markers_AM <- c("CD68","MARCO", "C1QC", "FABP4")
markers_mono <- c("CSF1R", "TMEM176B", "NR4A2")
markers_baso <- c("HDC", "MS4A2", "GATA2", "CPA3", "TPSAB1")
markers_CD8 <- c("CD8A", "CLIC3")
Markers_CD4 <- c("CD4", "IL7R")
Markers_eosinophile <- c("SIGLEC8", "SYNE1", "CLC")

DotPlot(Cell_HiveAnnotate, features = c(markers_AM,markers_mono, "VCAN","MAFB",
markers_neutro, "MKI67", "TOP2A",markers_CD8, Markers_CD4, Marker_NK,
Markers_eosinophile, marker_epi, markers_baso),
        group.by = "Annotation", cols = c("lightgray", "darkorange"),) + theme_classic()
        + scale_size(range = c(0,12)) + theme(axis.text.x = element_text(angle = 45,
        hjust = 1), axis.text = element_text(size = 35),
           legend.text   = element_text(size = 25), legend.key.height= unit(1, 'cm'),
        legend.key.width= unit(2, 'cm'))
```

```
## Scale for size is already present.
## Adding another scale for size, which will replace the existing scale.
```




```
#ggsave("Dotplot_annotated.pdf", width = 35, height = 10)
```

```
var_order <- c(1,4,5,6,7,8,9,10,11,2,3)
dittoBarPlot(Cell_HiveAnnotate, "Annotation", group.by ="orig.ident", var.labels.reorder
= var_order, color.panel = col)
```


Annotation
1.0
0.5
0.0
Percent of cells
Hive1
Hive2
Hive3
Hive4
orig.ident
C1 − AM
C2 − AM
C3 − AM
C4 − MAFB+ MoMac
C5 − Neu
C6 − Neu
C7 − T cell
C8 − T cell
C9 − Eos
C10 − Epithelial cell
C11 − Mast cell

```r
#saveRDS(Cell_HiveAnnotate, "lavage_Hive_annotated.rds")
```

```r
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] tidyHeatmap_1.8.1     plotly_4.10.2         ComplexHeatmap_2.16.0
##  [4] dittoSeq_1.12.0       RColorBrewer_1.1-3    ggplot2_3.4.2
##  [7] patchwork_1.1.2       SeuratObject_4.1.3    Seurat_4.3.0
## [10] dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21            splines_4.3.3
##   [3] later_1.3.1                 bitops_1.0-7
##   [5] tibble_3.2.1                polyclip_1.10-4
##   [7] lifecycle_1.0.3             doParallel_1.0.17
##   [9] globals_0.16.2              lattice_0.22-5
##  [11] MASS_7.3-60.0.1             dendextend_1.17.1
##  [13] magrittr_2.0.3              limma_3.56.2
##  [15] rmarkdown_2.23              yaml_2.3.7
##  [17] httpuv_1.6.11               sctransform_0.3.5
##  [19] spam_2.9-1                  sp_2.0-0
##  [21] spatstat.sparse_3.0-2       reticulate_1.30
##  [23] cowplot_1.1.1               pbapply_1.7-2
##  [25] abind_1.4-5                 zlibbioc_1.46.0
##  [27] Rtsne_0.16                  GenomicRanges_1.52.0
##  [29] purrr_1.0.1                 BiocGenerics_0.46.0
##  [31] RCurl_1.98-1.12             circlize_0.4.15
##  [33] GenomeInfoDbData_1.2.10     IRanges_2.34.0
##  [35] S4Vectors_0.38.1            ggrepel_0.9.3
##  [37] irlba_2.3.5.1               listenv_0.9.0
```

```
##  [39] spatstat.utils_3.0-3        pheatmap_1.0.12
##  [41] goftest_1.2-3               spatstat.random_3.1-5
##  [43] fitdistrplus_1.1-11         parallelly_1.36.0
##  [45] leiden_0.4.3                codetools_0.2-19
##  [47] DelayedArray_0.26.3         tidyselect_1.2.0
##  [49] shape_1.4.6                 farver_2.1.1
##  [51] viridis_0.6.3               matrixStats_1.0.0
##  [53] stats4_4.3.3                spatstat.explore_3.2-1
##  [55] jsonlite_1.8.7              GetoptLong_1.0.5
##  [57] ellipsis_0.3.2              progressr_0.13.0
##  [59] ggridges_0.5.4              survival_3.5-8
##  [61] iterators_1.0.14            foreach_1.5.2
##  [63] tools_4.3.3                 ica_1.0-3
##  [65] Rcpp_1.0.11                 glue_1.6.2
##  [67] gridExtra_2.3               xfun_0.39
##  [69] MatrixGenerics_1.12.2       GenomeInfoDb_1.36.0
##  [71] withr_2.5.0                 fastmap_1.1.1
##  [73] fansi_1.0.4                 digest_0.6.33
##  [75] R6_2.5.1                    mime_0.12
##  [77] colorspace_2.1-0            scattermore_1.2
##  [79] Cairo_1.6-2                 tensor_1.5
##  [81] spatstat.data_3.0-1         utf8_1.2.3
##  [83] tidyr_1.3.0                 generics_0.1.3
##  [85] data.table_1.14.8           httr_1.4.6
##  [87] htmlwidgets_1.6.2           S4Arrays_1.2.1
##  [89] uwot_0.1.16                 pkgconfig_2.0.3
##  [91] gtable_0.3.3                lmtest_0.9-40
##  [93] SingleCellExperiment_1.22.0 XVector_0.40.0
##  [95] htmltools_0.5.5             dotCall64_1.0-2
##  [97] clue_0.3-64                 scales_1.2.1
##  [99] Biobase_2.60.0              png_0.1-8
## [101] knitr_1.43                  rstudioapi_0.14
## [103] reshape2_1.4.4              rjson_0.2.21
## [105] nlme_3.1-164                zoo_1.8-12
## [107] GlobalOptions_0.1.2         stringr_1.5.0
## [109] KernSmooth_2.23-22          parallel_4.3.3
## [111] miniUI_0.1.1.1              pillar_1.9.0
## [113] vctrs_0.6.3                 RANN_2.6.1
## [115] promises_1.2.0.1            xtable_1.8-4
## [117] cluster_2.1.6               evaluate_0.21
## [119] magick_2.7.5                cli_3.6.1
## [121] compiler_4.3.3              rlang_1.1.1
## [123] crayon_1.5.2                future.apply_1.11.0
## [125] labeling_0.4.2              plyr_1.8.8
## [127] stringi_1.7.12              viridisLite_0.4.2
## [129] deldir_1.0-9                munsell_0.5.0
## [131] lazyeval_0.2.2              spatstat.geom_3.2-4
## [133] Matrix_1.6-1                future_1.33.0
## [135] shiny_1.7.4.1               SummarizedExperiment_1.30.2
## [137] highr_0.10                  ROCR_1.0-11
## [139] igraph_1.5.0.1
```